

Bank Customer Churn Prediction Using ANN

The objective was to develop a predictive model to identify the customers who are at high risk of churning, enabling proactive retention strategies by using the Artificial Neural Network.

Technical stack:

- Language: Python
- Machine learning techniques: Tenseflow/Keras
- Data Preprocessing: numpy, pandas, Scikit-learn
- Visualization: Matplotlib, seaborn

Methodology:

1) Data Preprocessing :

Performed a comprehensive data cleaning ,which including encoding the variables using labelencoder and normalizing the features using standardscaler to stabilize gradient descent during learning.

2) ANN Model Design:

Constructed multilayer feedforward Neural Network

- The input layer is adjusted to match the dimensionality of the feature set.
- Hidden layers uses dense layers with ReLu activation function, so that it can effectively capture the complex, non-linear relationships within the dataset.
- Output Layer ,deployed the sigmoid activation function to generate the probability prediction .i.e 1 = stay and 0 = churn

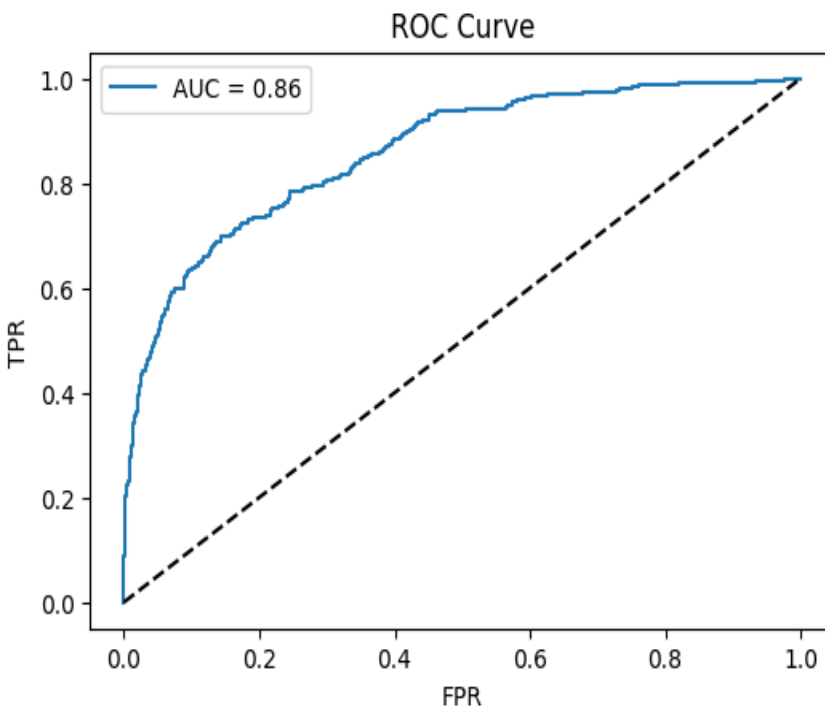
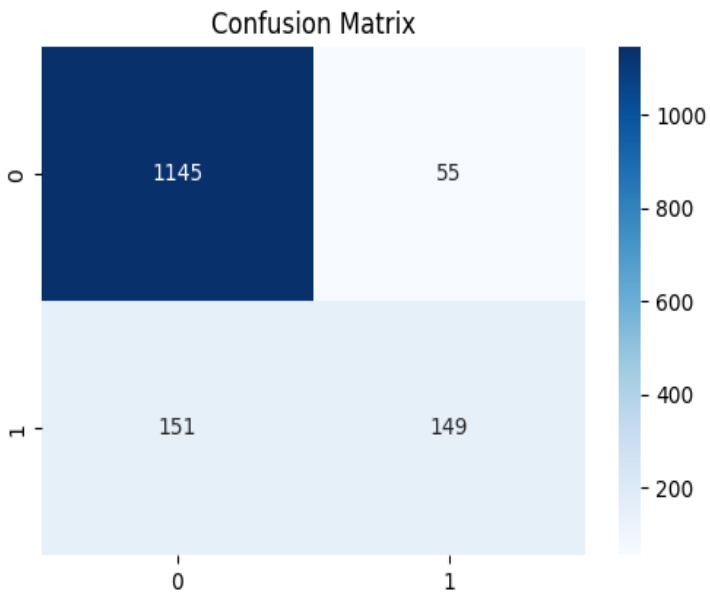
3) Model Training & Evaluation:

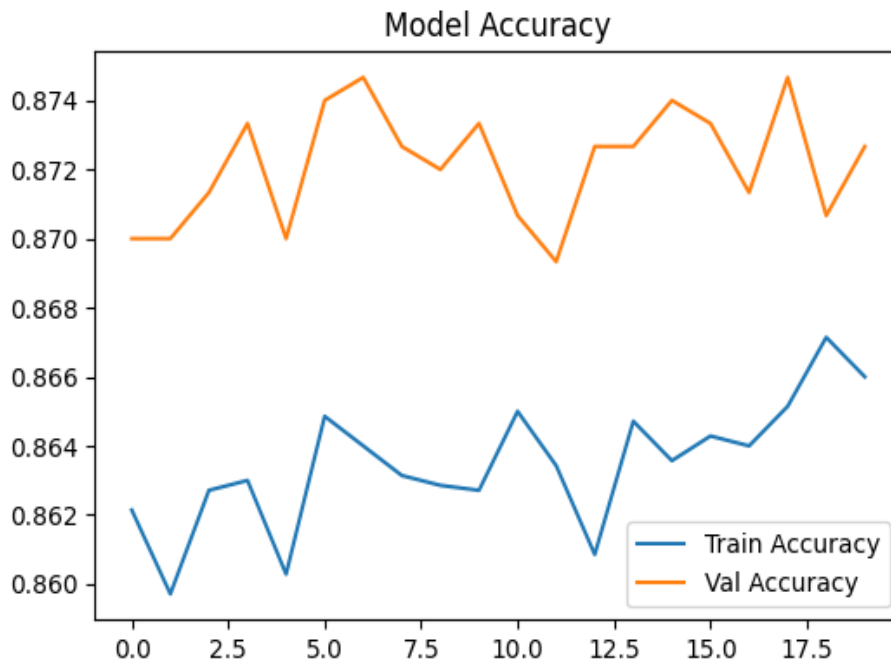
- Loss Function: Binary Cross-Entropy.
- Optimizer: Adam.
- Performance Metrics: Evaluated performance using the Confusion Matrix to track True/False Positives/Negatives and the ROC-AUC score to measure the model's discriminative power.

Key Findings

Performance: Mention the 86% AUC score. Explain that this demonstrates the model's robustness in correctly identifying potential churners.

Visualization:





"The learning curves show stable convergence with minimal gap between training and validation accuracy, indicating effective generalization."